

HKUST MATH 2431 problem set 3

1.

1. (*) The Monty Hall problem

[4 Points]

In a game show, you have to choose one of three doors. Behind one door is a new car, and behind the other two doors are goats. After you make your choice, the host reveals one of the other doors, which always has a goat behind it. You are then given the option to switch your choice to the remaining unopened door.

- (a) Given that the host has opened a door revealing a goat, should you switch your choice to increase the chances of winning the car? Please explain your reasoning.

1. a. yes. Analyze by splitting cases.

1. Orig choice has car $\rightarrow$ Switching means no car

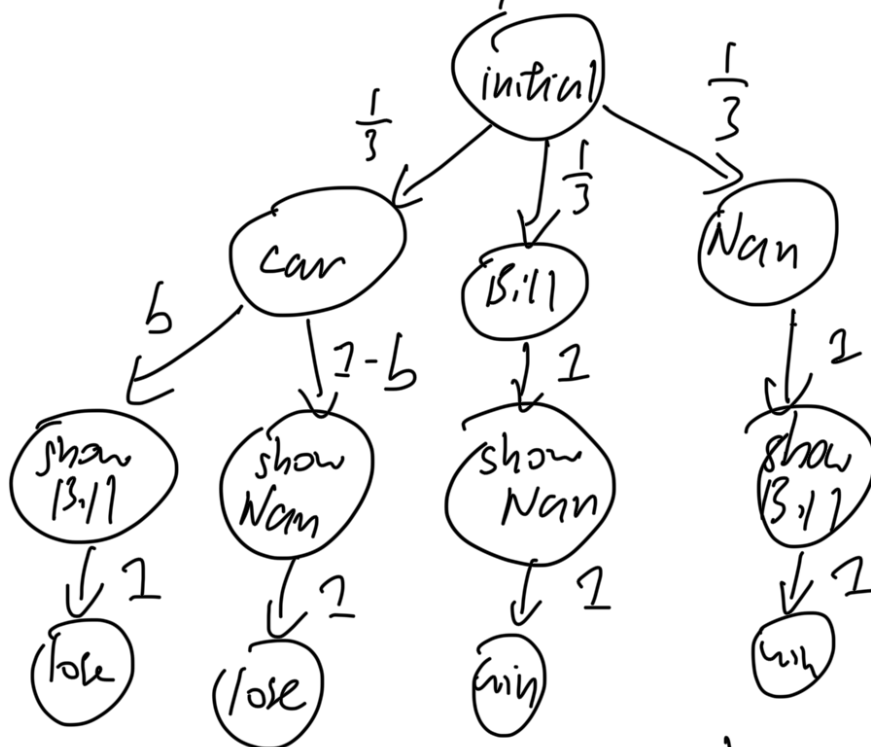
2. Orig choice had no car $\rightarrow$ only 1 door with goat, the other door is a car, so the door not open by the host must be a car $\rightarrow$ switching means car, always

If not switch, then $1/3$ chance of case 1, which yields car when not switch; case 2 yields no car.
If switch, then $2/3$ chance of case 2, which yields car when switch; case 1 yields no car.

So switching is better.

- (b) Let's call the two goats Bill and Nan. Given that the host has opened a door revealing a goat, and shows Bill with a probability of b , following the strategy in (a), what is the probability of winning the car, *i.e.* winning a car given Bill is revealed? [Note: This means that if the host can reveal either Bill or Nan (a door with the car behind it is initially chosen), the host will reveal Bill with probability b .]

1. b, Draw a probability tree when the strat is always switch



$$P(\text{win} \mid \text{show Bill}) = \frac{\frac{1}{3}}{\frac{1}{3} \cdot b + \frac{1}{3}} = \frac{1}{1+b}$$

- (c) If the host opens a door randomly, which strategy would you take to maximize your chances of winning the car? [Hint: Consider two cases separately, one of which is trivial.]

1. (c) Now that the host opens a door randomly, it means the host's choice is independent of your first choice, possibly as well ignore the first choice at all since
- only the last choice matters for winning
 - the first choice does not affect the host's choice, and the first choice itself does not reveal any info for helping to win; that is, the first choice provides no info for helping to win at all.

Then the strategy is a function of the host's choice only, and the strategy is obvious:

- first choice: always ignored (just randomly choose one)
- host reveals his/her choice
 - If the host reveals a car, make it the final choice (switching from the first choice if the revealed car is not already chosen)
 - If the host reveals a goat,

make any of the other two the final choice
(switching from the first choice if already chosen the revealed goat)

3.

3. (*) **Discrete distributions**

[4 Points]

For (a) and (b), first determine which probability distribution you are using.

- (a) We roll a fair die until the first time we observe the number 6. What is the probability that we need more than 9 attempts?

3. a. Use geometric distribution

with $p = \frac{1}{6}$

and support on $\{0, 1, 2, \dots\}$

(there is another variant of geo dist
with support on $\{1, 2, 3, \dots\}$)

Then $P(k) = (1-p)^k p$

Need > 9 attempts $\Rightarrow$ 9 failures or more
 $\Rightarrow k \geq 9$

$$\begin{aligned} \text{req. prob} &= \sum_{k=9}^{\infty} (1-p)^k p \\ &= (1-p)^9 \sum_{k=0}^{\infty} (1-p)^k \\ &= (1-p)^9 \cdot p \cdot \frac{1}{p} \\ &= (1-p)^9 = \left(\frac{5}{6}\right)^9 \end{aligned}$$

- (b) In the production of a certain kind of computer chip defects occur independently with probability 0.01. How likely is it that there are at least 2 defects in a batch of 100 computer chips? Give the exact probability and compare it to the approximation with a Poisson distribution.

3.b. Use binomial distribution
with $p = 0.01$, $n = 100$

support is $\{0, \dots, 100\}$

$$P(k) = \binom{100}{k} (0.01)^k (0.99)^{100-k}$$

at least 2 defects $\Rightarrow 2 \leq k \leq 100$

$\Rightarrow$ complement of $0 \leq k \leq 1$

$$\begin{aligned} \text{req. prob.} &= 1 - \binom{100}{0} (0.01)^0 (0.99)^{100} \\ &\quad - \binom{100}{1} (0.01)^1 (0.99)^{99} \\ &= 1 - (1)(1)(0.99)^{100} \\ &\quad - (100)(0.01)(0.99)^{99} \\ &= 1 - 0.99^{100} - 0.99^{99} \\ &\approx 0.264238 \end{aligned}$$

For approx: use poisson distribution

with $\lambda = np = (100)(0.01) = 1$

support is $\{0, \dots, \infty\}$

$$P(k) = e^{-\lambda} \frac{\lambda^k}{k!} = \frac{1}{e} \cdot \frac{1}{k!}$$

$$\begin{aligned} \text{req. prob.} &= 1 - \left(\frac{1}{e}\right)\left(\frac{1}{1}\right) - \left(\frac{1}{e}\right)\left(\frac{1}{1}\right) \\ &= 1 - \frac{2}{e} \approx 0.264242 \end{aligned}$$

The approximation is quite accurate, only

differing starting from the 5th d.p.

It shows when n is large (100 is somewhat large), the approximation is highly accurate

Show that the following approximation of probability mass functions holds:

- (c) The probability mass function of a hypergeometric distribution $H(N, M, n)$ with parameters $N, M, n \in \mathbb{N}$ converges to the probability mass function of a binomial distribution $Bin(n, p)$ as $N, M \rightarrow \infty$ if $\frac{M}{N} \rightarrow p$.

3. c. For hypergeometric dist. $H(N, M, n)$

$$P_1(m) = \frac{\binom{M}{m} \binom{N-M}{n-m}}{\binom{N}{n}}$$

For binomial dist. $B_n(n, p)$

$$P_2(m) = \binom{n}{m} p^m (1-p)^{n-m}$$

$$P_2(m) = \frac{\binom{M}{m} \binom{N-M}{n-m}}{\binom{N}{n}}$$

$$= \frac{M!}{m! (M-m)!} \cdot \frac{(N-M)!}{(n-m)! (N-M)-(n-m)!} \cdot \frac{n! (N-n)!}{N!}$$

$$= \frac{n!}{m! (n-m)!} \cdot \frac{M! (N-M)! (N-n)!}{((M-m)! ((N-M)-(n-m))! N!}$$

$$= \binom{n}{m} \frac{M^m (N-M)^{n-m}}{N^n}$$

$$= \binom{n}{m} \frac{M^m (N-M)^{n-m}}{(N-m)^m N^{n-m}}$$

$$\sim \binom{n}{m} p^m (1-p)^{n-m}$$

$a^{\underline{b}}$ denote
falling factorial

For finite fixed
 n, m

as $N, M \rightarrow \infty$

$$\frac{M^m}{(N-m)^m} \sim \left(\frac{M}{N}\right)^m \sim p^m$$

$$\frac{(N-M)^{n-m}}{N^{n-m}} \sim \left(\frac{N-M}{N}\right)^{n-m}$$

$$\sim (1-p)^{n-n}$$

Consider the following thinning of a Poisson distribution:

- (d) Assume that the number of penalty kicks during a football match is Poisson distributed with some parameter $\lambda > 0$. Also assume that every penalty kick results in a goal with the same probability $p \in (0, 1)$, independently of all others. Determine the distribution of the number of goals obtained via a penalty kick.

Hint: For $k \in \mathbb{N}_0$, calculate the probability that there are exactly k goals obtained via a penalty kick.

3. d. The dist. has a part of

$$\begin{aligned}
 P(m) &= \sum_{k=m}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} \binom{k}{m} p^m (1-p)^{k-m} \\
 &= \sum_{k=m}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} \frac{k!}{(k-m)! m!} p^m (1-p)^{k-m} \\
 &= \sum_{k=m}^{\infty} e^{-\lambda} \lambda^k \frac{(1-p)^{k-m}}{(k-m)!} \frac{p^m}{m!} \\
 &= \frac{p^m}{m!} \lambda^m e^{-\lambda p} \sum_{k=m}^{\infty} e^{-\lambda(1-p)} \lambda^{k-m} \frac{(1-p)^{k-m}}{(k-m)!} \\
 &= \underbrace{\left(\frac{(\lambda p)^m}{m!} e^{-\lambda p} \right)}_{\substack{\text{part of} \\ \text{Poisson}(\lambda p)}} \underbrace{\sum_{k=0}^{\infty} e^{-\lambda(1-p)} \frac{(\lambda(1-p))^{k-m}}{(k-m)!}}_{\substack{\text{Poisson}(\lambda(1-p)) \\ \text{this equals 1}}}
 \end{aligned}$$

So our dist. is just $\text{Poisson}(\lambda p)$.
 Intuitively, this makes sense.